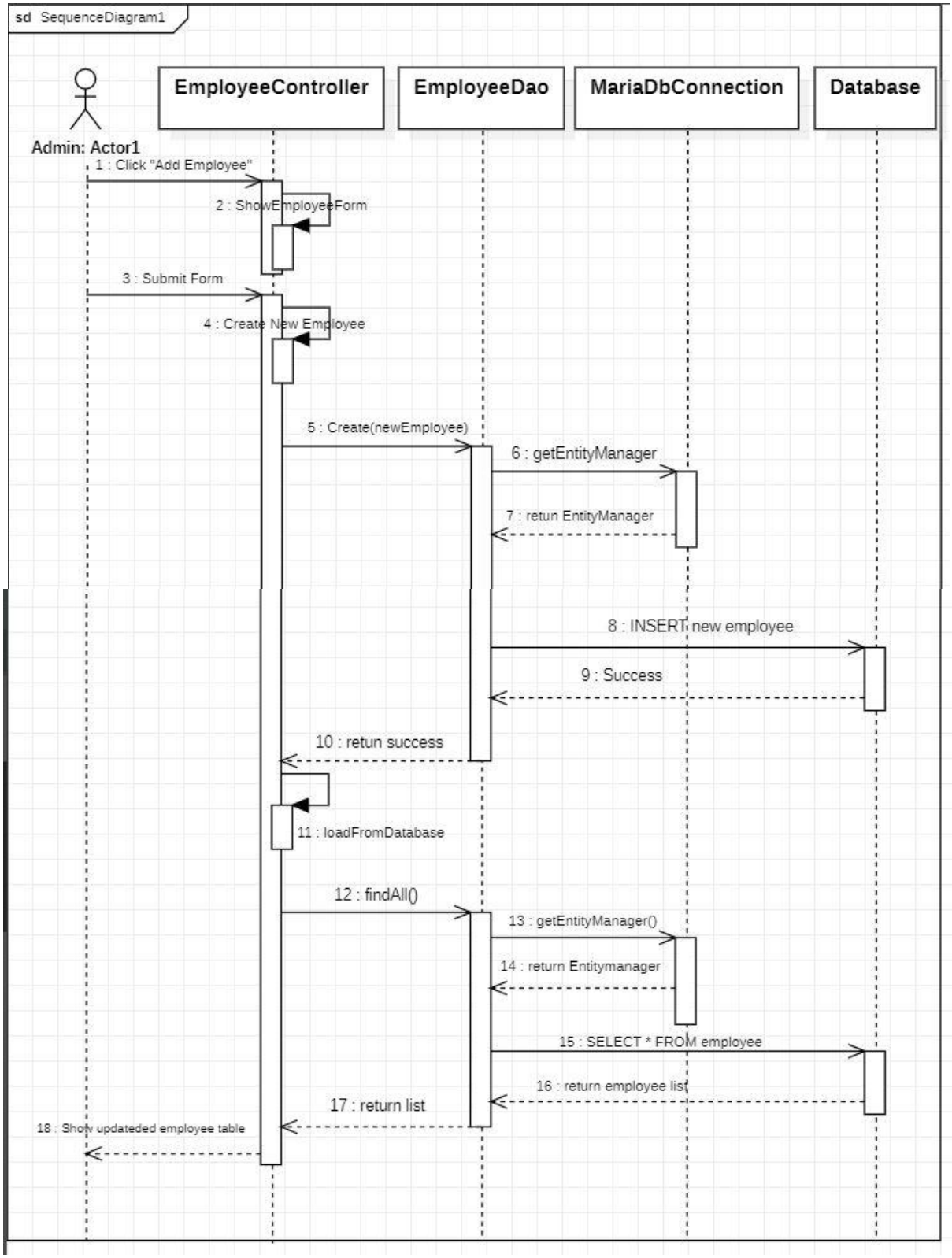
Sequence Diagram



Description for the Add Employee Sequence Diagram

1. Admin opens the Add Employee form

The Admin clicks the “Add Employee” button in the Employee Management view. The EmployeeController responds by opening a popup form where the Admin can enter the new employee’s details.

2. Admin submits the employee form

After filling in the required fields (name, gender, department, contract type, etc.), the Admin clicks the Submit button. The form data is collected inside the controller.

3. EmployeeController creates a new Employee object

The controller constructs a new Employee entity using the values entered in the form. This object represents the employee that will be stored in the database.

4. EmployeeController calls EmployeeDao.create()

The controller sends the new Employee object to the EmployeeDao by calling the create(newEmp) method. This begins the persistence process.

5. EmployeeDao requests an EntityManager

Inside the DAO, the method calls MariaDbConnection.getEntityManager() to obtain an EntityManager instance. This object manages the connection to the MariaDB database.

6. EmployeeDao persists the Employee

The DAO uses JPA to begin a transaction, persist the Employee entity, and commit the transaction. This inserts a new row into the employee table.

7. EntityManager is closed

After the operation, the DAO closes the EntityManager to free database resources.

8. EmployeeDao returns control to the controller

Once the employee is successfully saved, the DAO returns to the controller.

9. EmployeeController reloads the employee table

The controller calls loadFromDatabase(), which retrieves all employees using employeeDao.findAll() and refreshes the table in the UI.

10. Admin sees a success result

The popup closes, and the Admin sees the updated employee list with the newly added employee included.